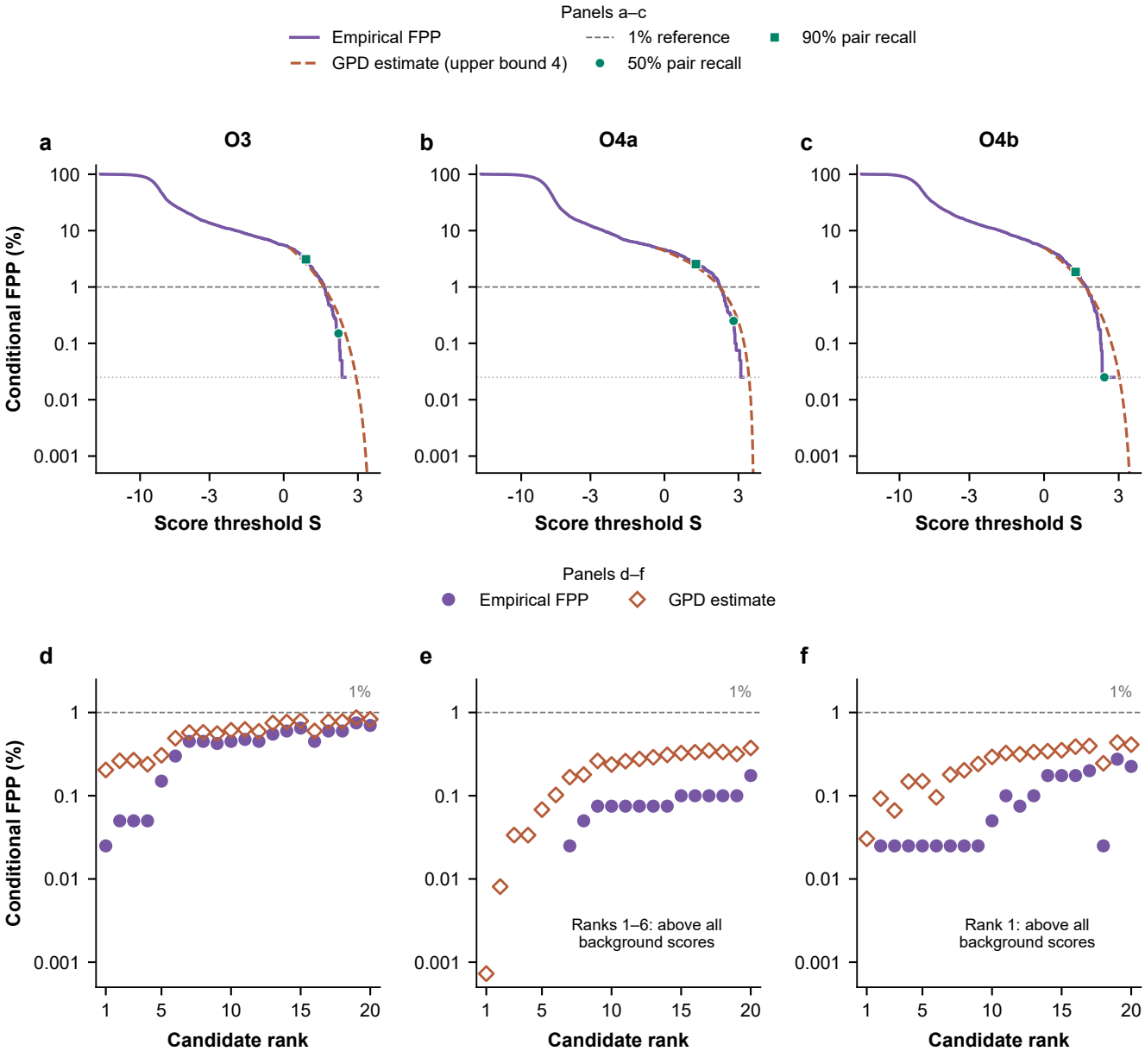

Panels a–c
Empirical FPP
GPD estimate (upper bound 4)
1% reference
50% pair recall
90% pair recall
a
O3
b
O4a
c
O4b
Conditional FPP (%)
100
10
1
0.1
0.01
0.001
-10
-3
0
3
Score threshold S
Panels d–f
Empirical FPP
GPD estimate
d
e
f
1%
Ranks 1–6: above all background scores
Rank 1: above all background scores
1
5
10
15
20
Candidate rank